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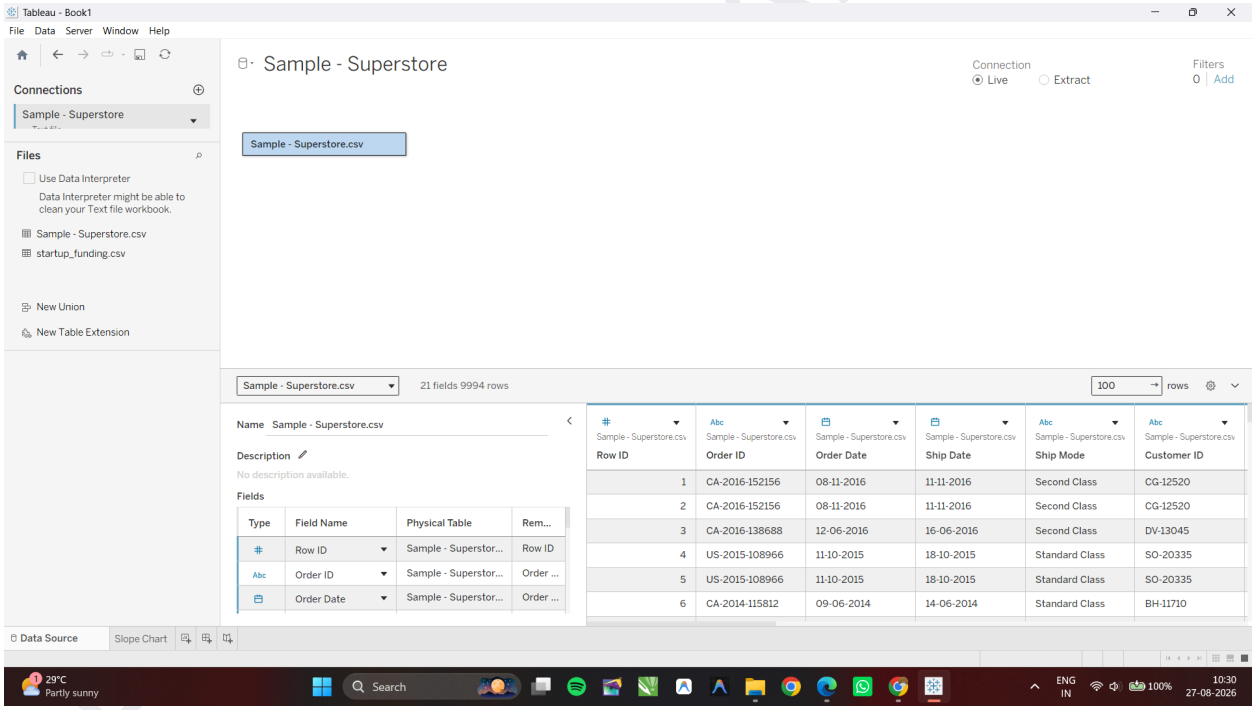
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Aim: Implementing Advanced Visualization Techniques and Customizations.
a) Create slope charts and lollipop charts
b) Design waterfall charts
c) Build sparklines for trend analysis
d) Implement sheet swapping for dynamic dashboards
e) Create custom geographic territories
f) Use background images in dashboards
g) Apply animation and transparency techniques

A: Create Slope Charts and Lollipop Charts
Task 1: Create a Slope Chart (Year-over-Year Progression)

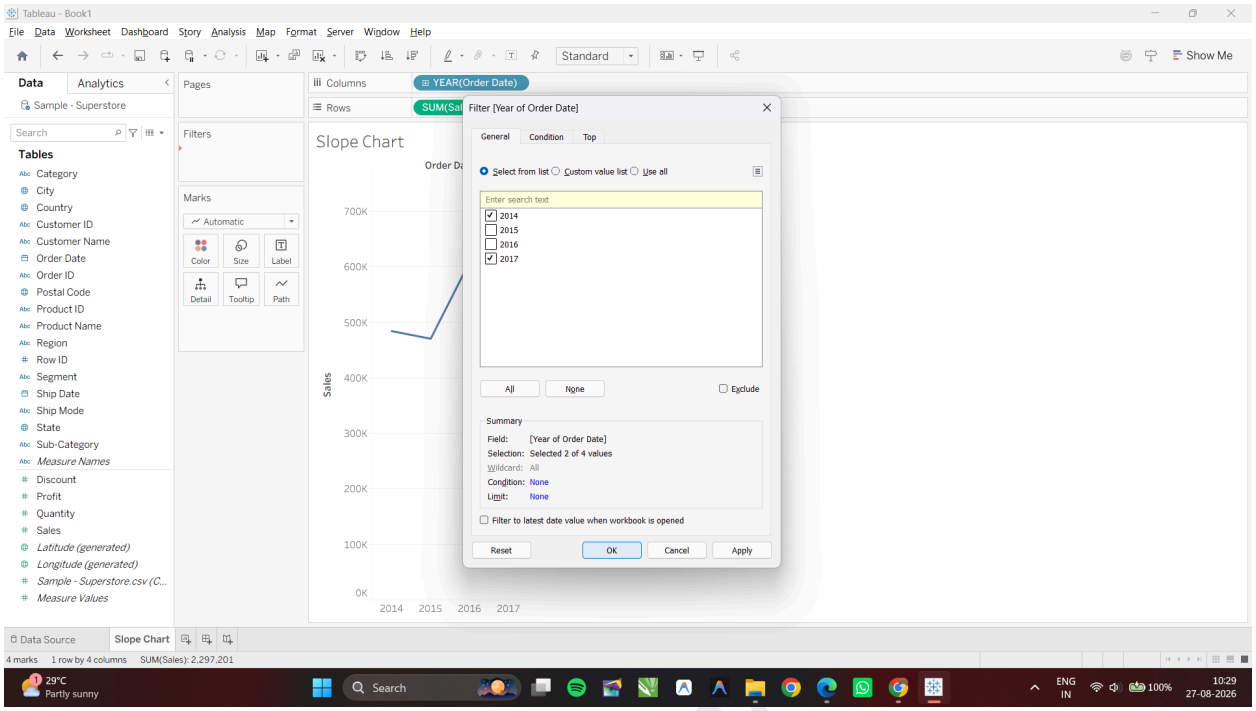
Create a **Slope Chart** by plotting **SUM(Sales)** for **2014 and 2017** using **YEAR(Order Date)**, adding **Sub-Category** to Detail and Color, then enabling labels for both ends of each line.

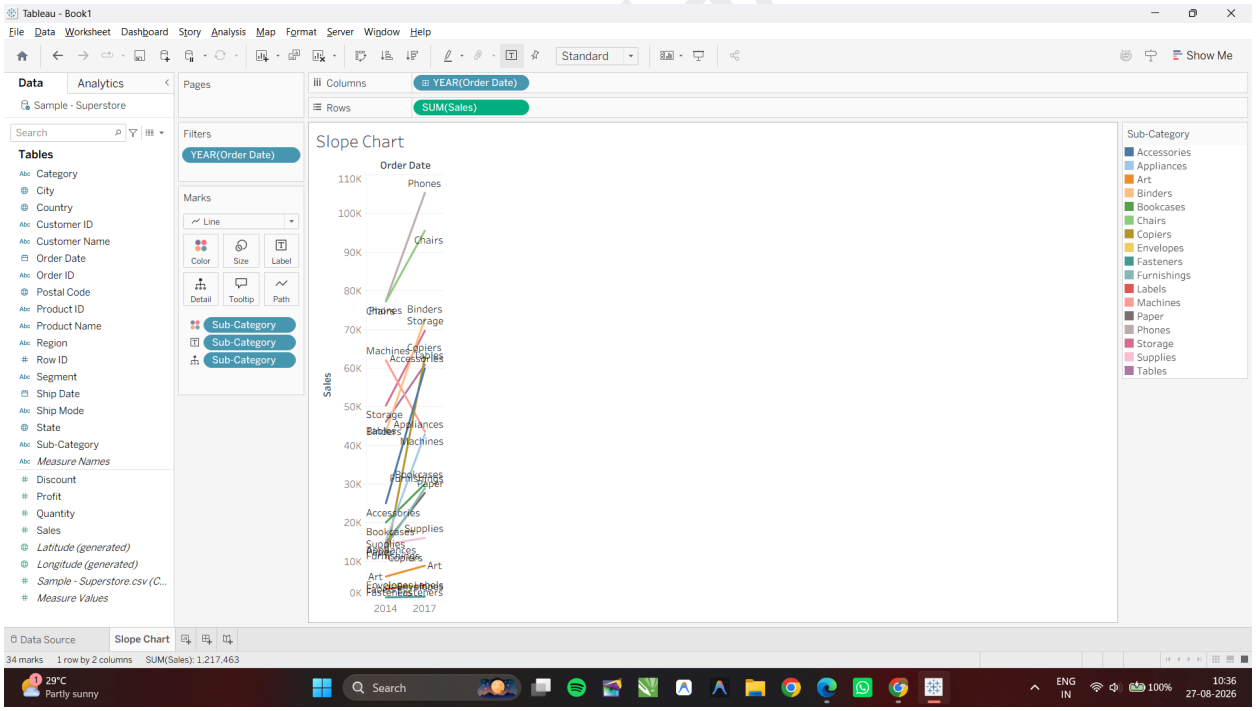


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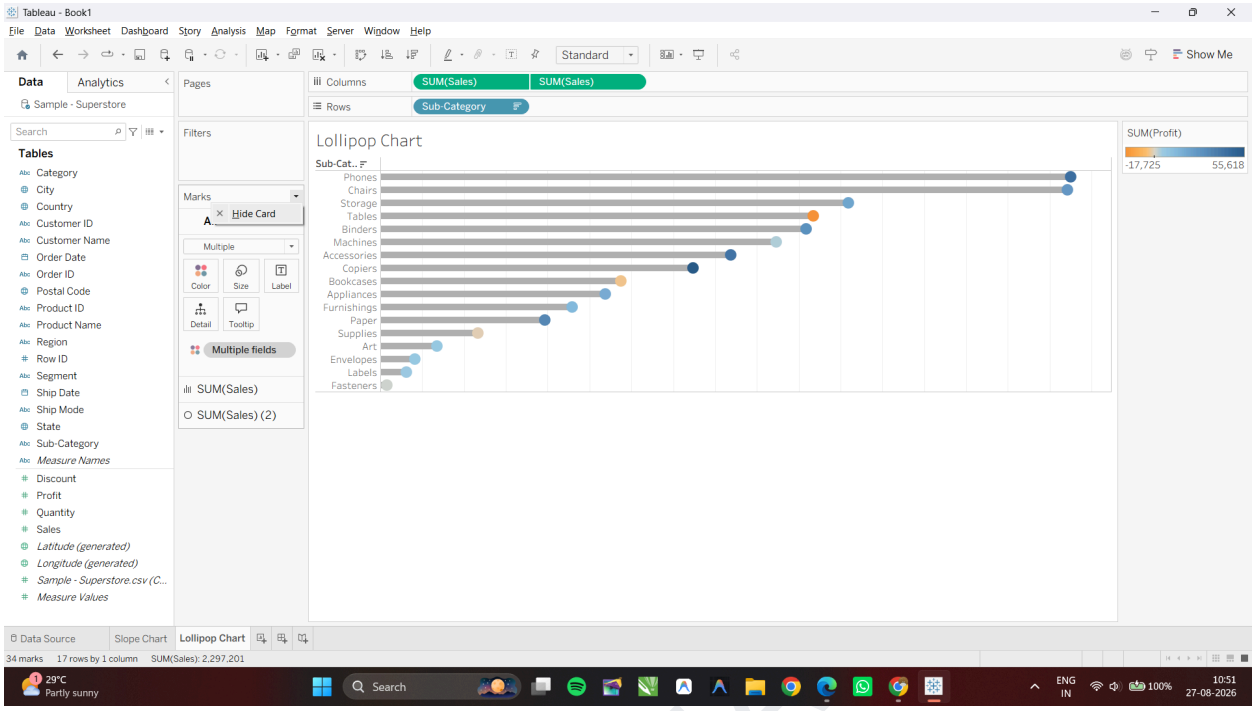


Task 2: Create a Lollipop Chart (Sales with Profit Highlighting)
Create a **Lollipop Chart** by placing **Sub-Category** on Rows and **Sales twice** on Columns, converting them to a **Dual Axis**, using a thin **Bar** for the stem and a **Circle colored by Profit** for the head, then hiding the extra header and sorting by Sales.

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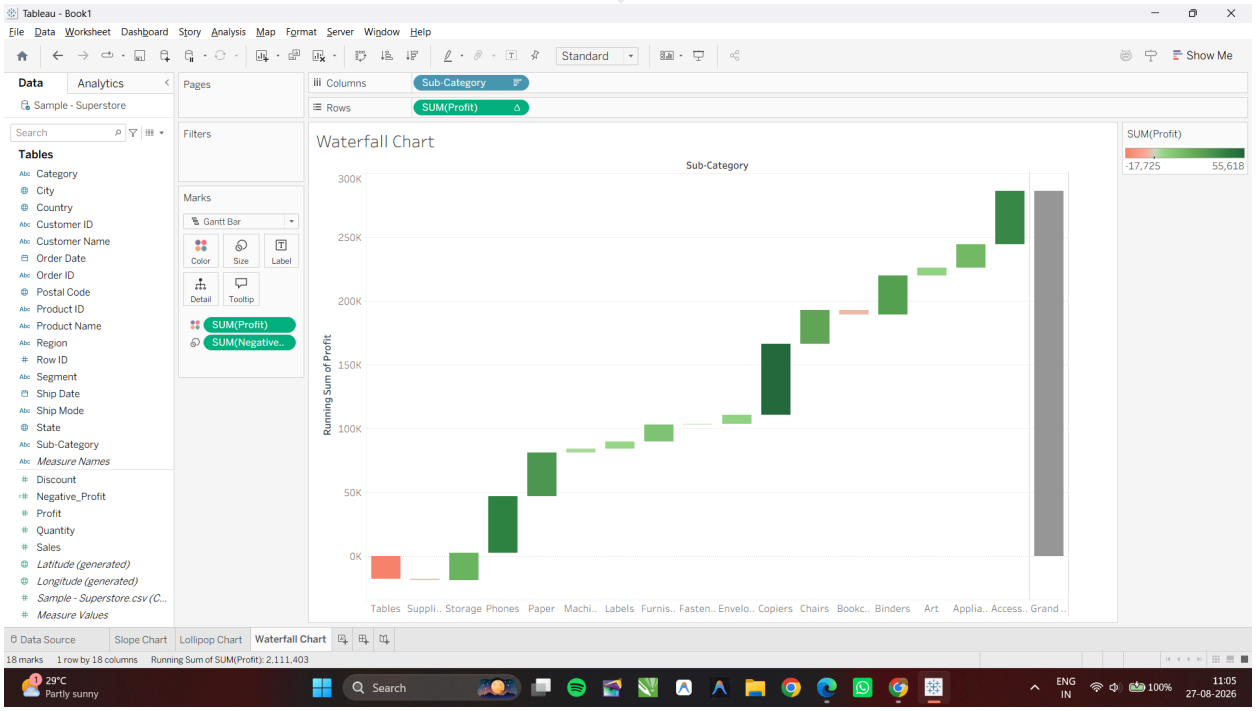
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B: Design Waterfall Charts

Create a **Waterfall Chart** by plotting **Sub-Category vs. Profit**, applying a **Running Total**, changing the mark type to **Gantt Bar**, using **Negative_Profit (-[Profit])** on Size and **Profit** on Color, then adding the **Grand Total**.



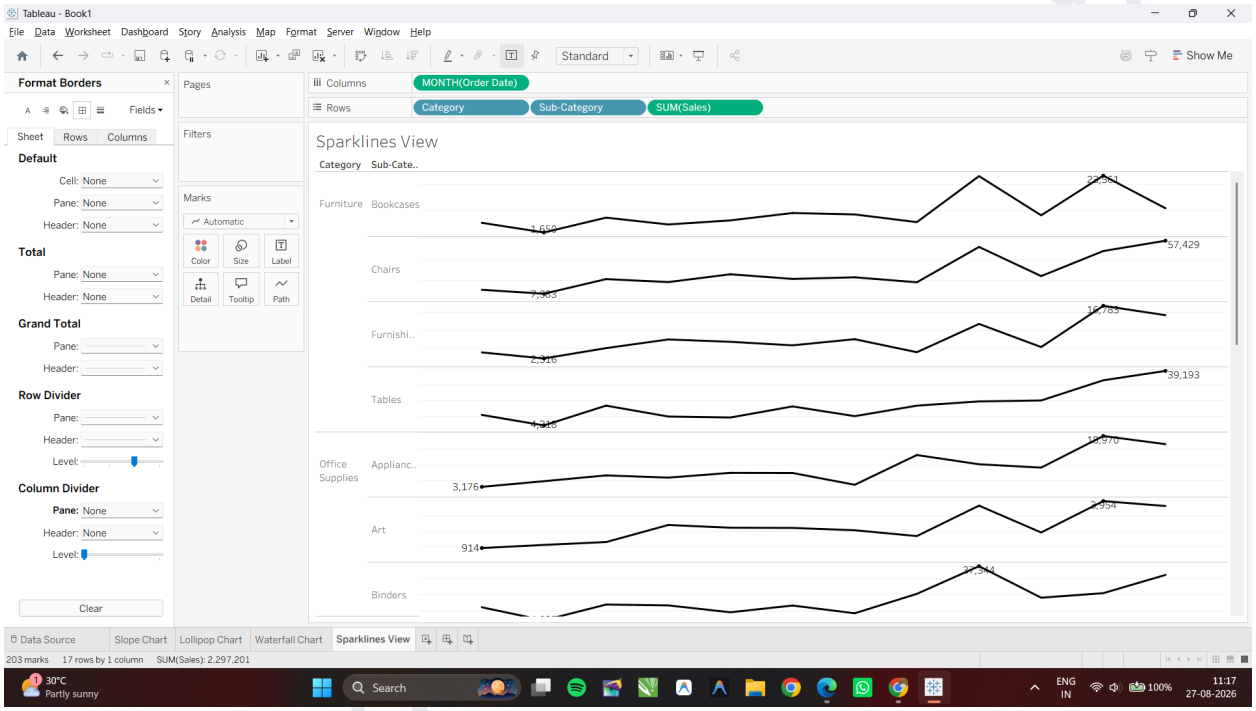
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C: Build Sparklines for Trend Analysis

Create a **Sparklines View** by placing **Category** and **Sub-Category** on Rows, **continuous MONTH(Order Date)** on Columns, and **Sales** on Rows to display individual time-series trends. Set independent axis ranges for each row, hide both axes and unnecessary gridlines, add subtle row dividers, and use **Min/Max labels** on **SUM(Sales)** to highlight the minimum and maximum values for each sparkline.



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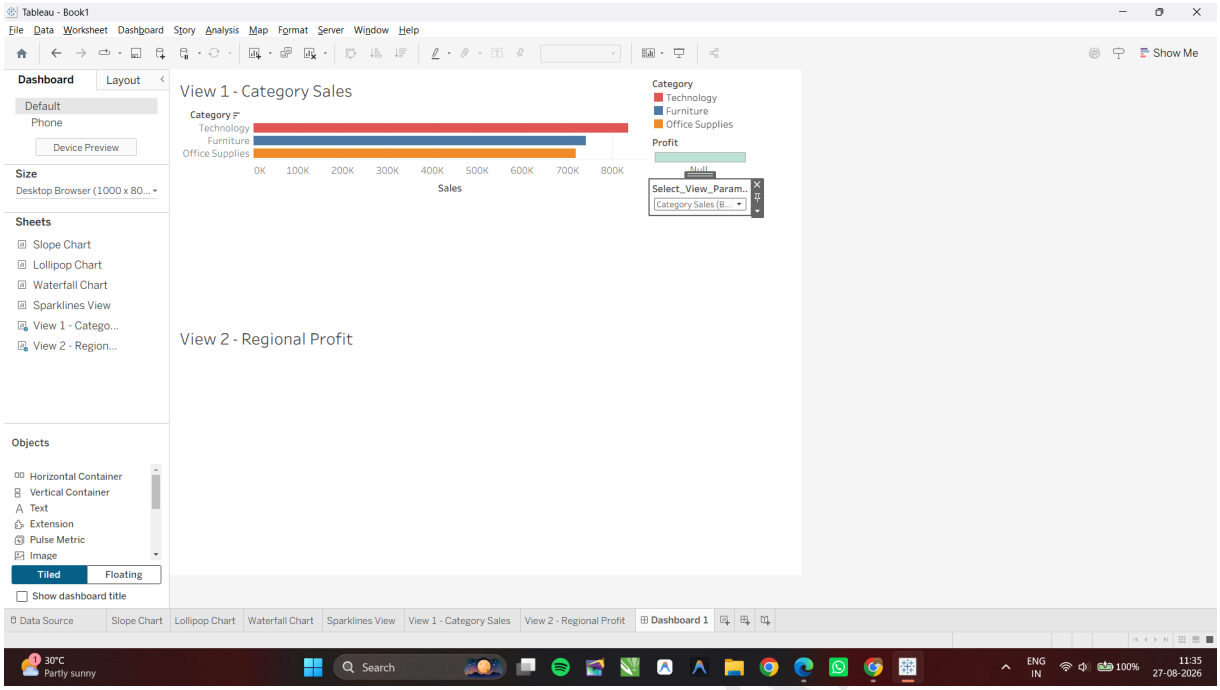
D: Execute Cross-Database Joins

Create **Select_View_Parameter** and **View_Filter**, build the **Category Sales** and **Regional Profit** views, then use the parameter dropdown to dynamically switch between both sheets on the dashboard.

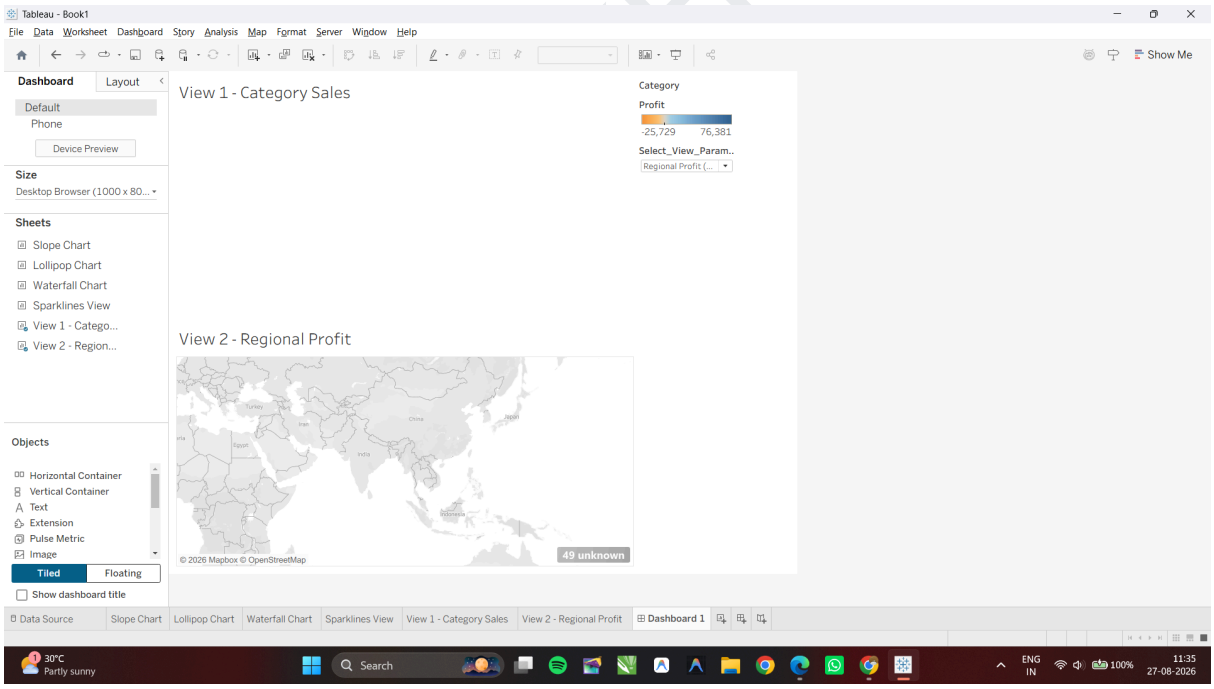
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Select **Regional Profit (Map)**: Sheet 1 collapses completely, and Sheet 2 expands to occupy the container.



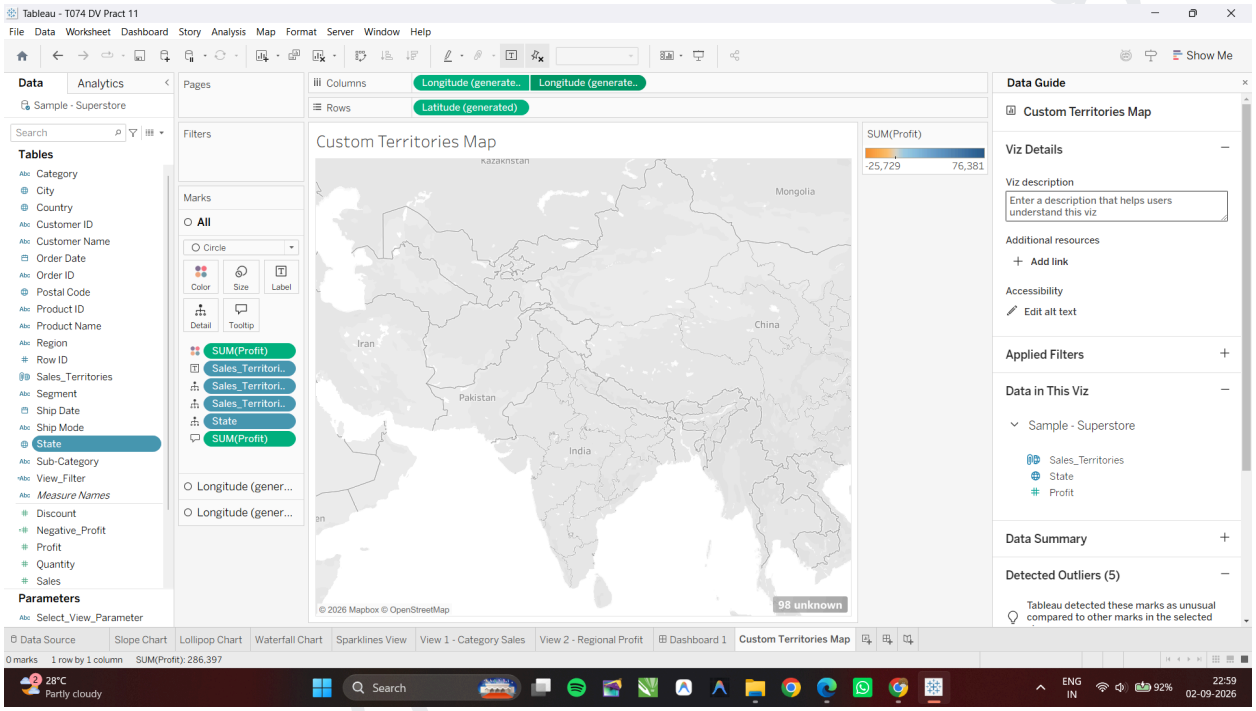
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E: Create Custom Geographic Territories

Create a **Custom Territories Map** by grouping the **State** field into four sales territories—**Pacific & Mountain Zone, Midwest & Central Hub, Atlantic & North-East, and Southern Enterprise Territory**—and rename the group field **Sales_Territories**. Build a map using the territory groups with **Sales on Color, Profit on Tooltip, and territory names on Label**, then optionally add a second **State-level Circle layer** using a **Dual Axis** to compare individual states with the custom territory view.



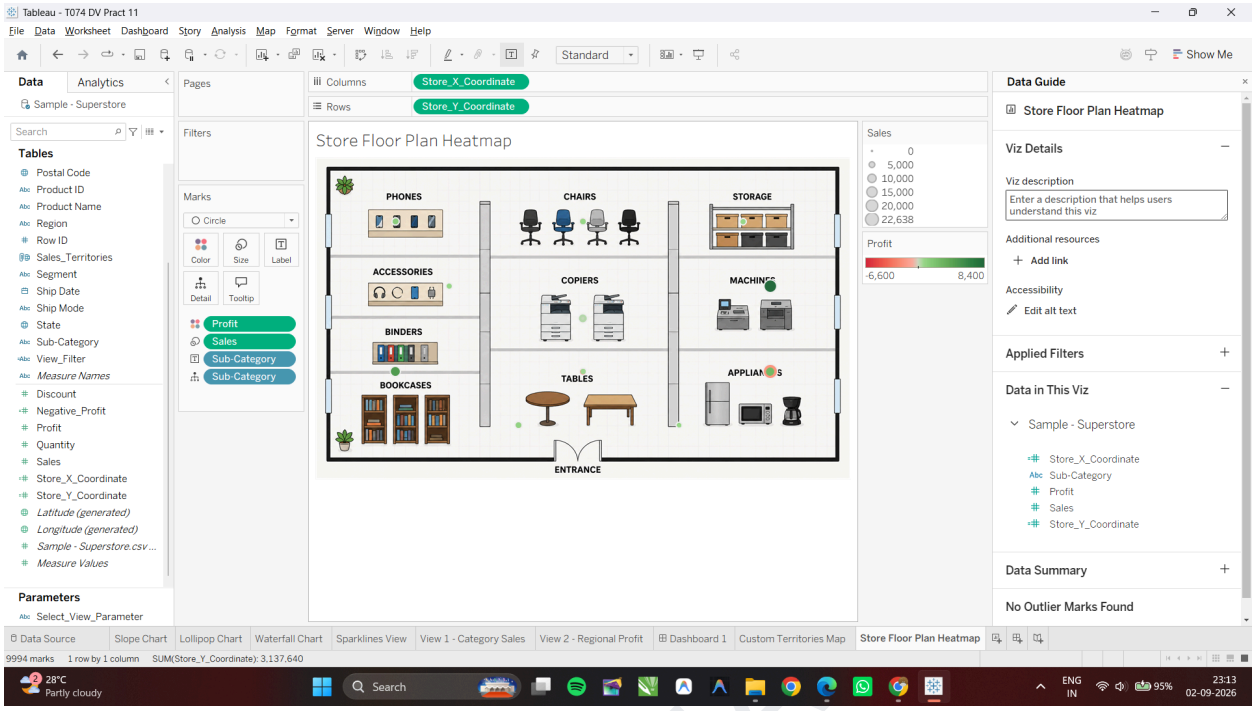
F: Use Background Images in Dashboards

Create a **Store Floor Plan Heatmap** by making **Store_X_Coordinate** and **Store_Y_Coordinate** calculated fields based on each **Sub-Category**, then add a floor-plan image as a background with the coordinate ranges **X: 0–1000** and **Y: 0–600**. Plot the coordinates using **Circle** marks, add **Sub-Category** to Detail and Label, use **Sales** for Size and **Profit** for Color with a **Red-Green Diverging** palette, then hide the axes and gridlines for a clean floor-plan visualization.

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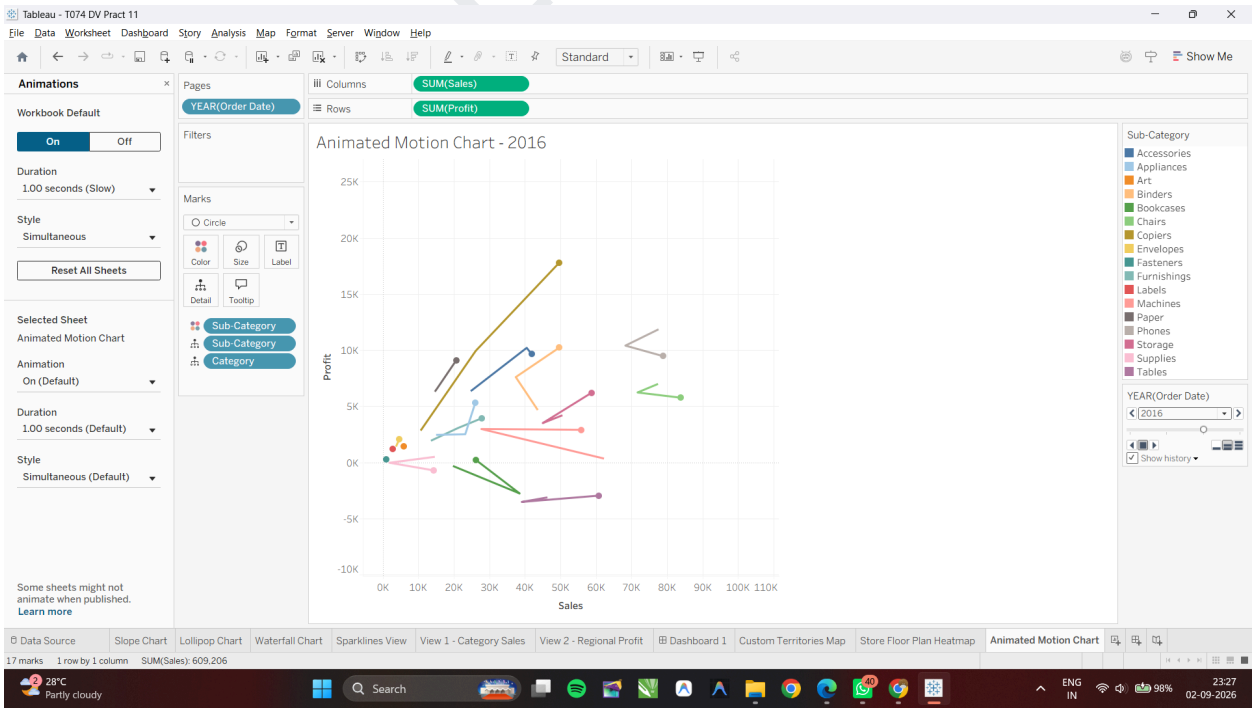
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G: Apply Advanced Data Cleaning Techniques

Task 1: Create Animated Motion Chart

Create an **Animated Motion Chart** by plotting **Sales vs Profit** as circles, using **Sub-Category** for Detail and Color and optionally **Category** for Shape. Add **YEAR(Order Date)** to the **Pages shelf**, enable **Show History** with marks and trails, adjust the playback speed, and turn on **Workbook Animations** to visualize how Sales and Profit change over time.



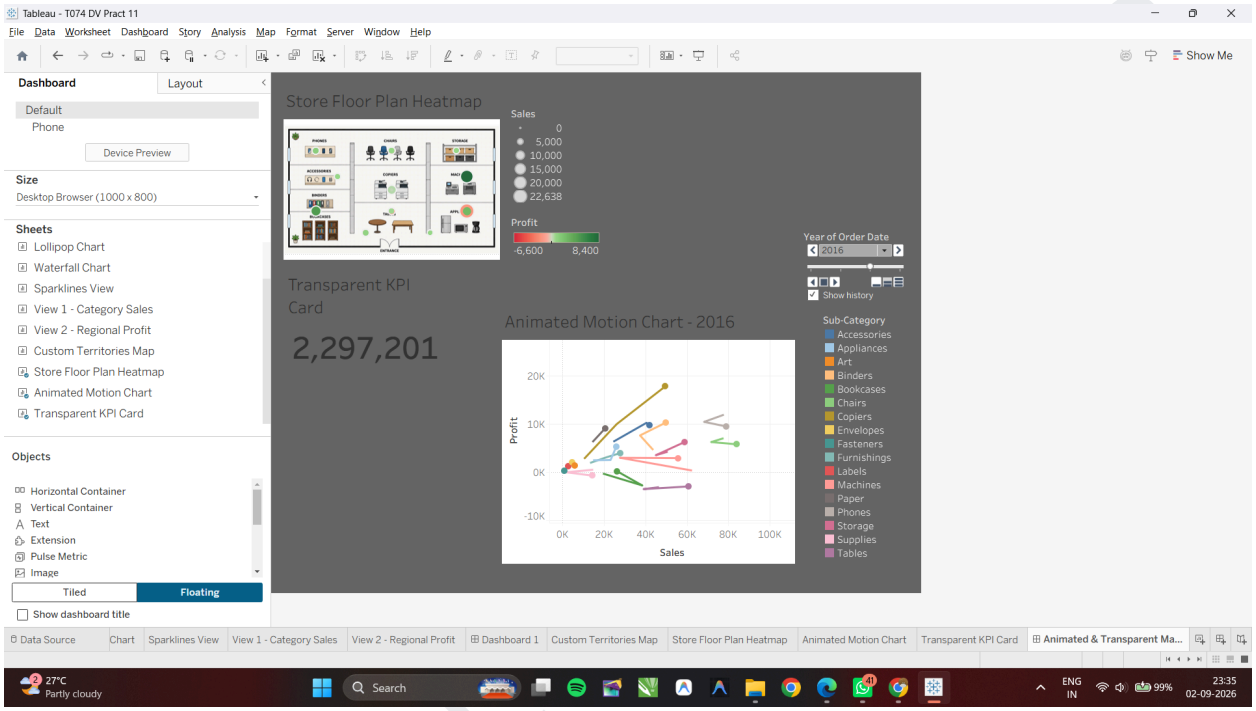
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Task 2: Transparent KPI Card

Create the **Animated & Transparent Master View** dashboard with a dark background, use **Floating** objects instead of Tiled, and place the **Store Floor Plan Heatmap** (or Custom Territories Map) as the main background. Overlay a **Transparent KPI Card** and the **Animated Motion Chart** as floating elements, resizing and positioning them to create a layered interactive dashboard.



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